以这些方式行走：利用行为多样性调优机器人控制以实现泛化

Gabriel B. Margolis、Pulkit Agrawal，Improbable AI 实验室，麻省理工学院{gmargo，pulkitag}@mit.edu

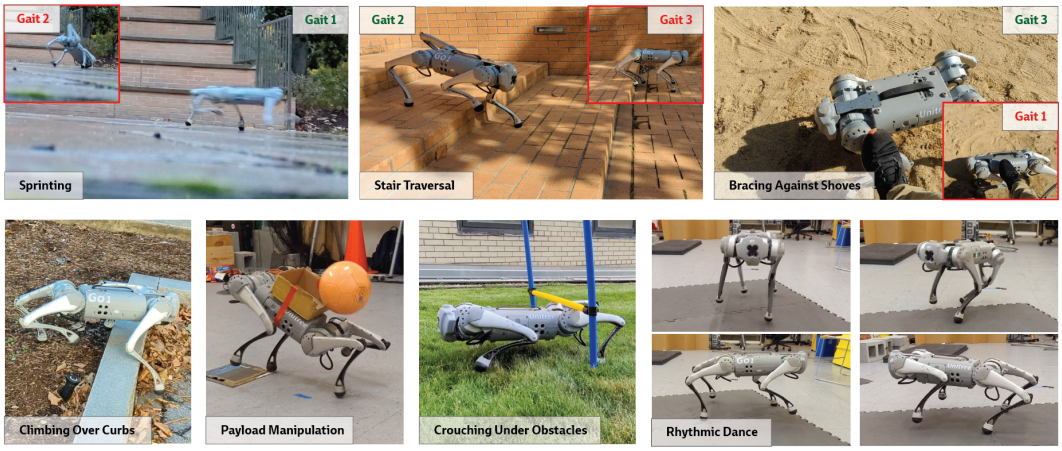


图 1：行为多样性（MoB）使人类能够将在平地上训练出的单个四足策略调整到各种未见过的环境中。第一行：低频步态在湿滑地形上无法冲刺（步态 2；插图），但将其调整为高频后便成功（步态 1）。然而，爬楼梯需要低频和高抬脚高度（步态 2；中间图像）。低抬脚和宽站姿（步态 3）使机器人能抵御腿部推挤，但冲刺成功的步态 1 却会失败。因此，调整步态有助于泛化到不同任务。第二行：我们的控制器实现的其他行为示例。

摘要：习得的运动策略可以快速适应与训练时相似的多样化环境，但在分布外测试环境中失败时缺乏快速调节的机制。这导致在新任务上取得良好性能需要缓慢且迭代的奖励与环境重设计循环。作为替代方案，我们提出学习一个单一策略，该策略编码一组结构化的运动策略族，以不同方式解决训练任务，从而产生行为多样性（Multiplicity of Behavior，MoB）。不同的策略具有不同的泛化能力，可以针对新任务或环境实时选择，从而无需耗时的重新训练。我们发布了一个快速、鲁棒的开源 MoB 运动控制器 Walk These Ways，它能够以可变的摆腿、姿态和速度执行多种步态，解锁多种下游任务：蹲伏、跳跃、高速奔跑、爬楼梯、抵御推搡、节奏舞蹈等。视频和代码发布：https://gmargo11.github.io/walk-these-ways/

关键词：*L* ocomotion、强化学习、任务Sp ecification

1 引言

近期研究已证明，在仿真中通过强化学习训练的四足运动控制器能够成功迁移，从而穿越具有挑战性的自然地形 [1, 2,

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3]. 对多样化地形的适应是通过从感官观测中估计地形属性来实现的，控制器随后会利用这些属性（即在线系统辨识）。这一范式的成功依赖于两个假设：一是需要对部署期间可能变化的环境参数进行先验建模，二是需要能够从感官观测中估计这些参数。为了绕过第一个假设，一种可能的做法是在训练过程中广泛变化一大组环境参数。然而，这会因产生具有挑战性或不可行的运动场景而导致学习问题变得困难。为简化学习，设计者通常会选择一组参数，并在经过仔细限制的范围内进行随机化。即使在这种设置下，学习课程和奖励塑形等额外措施对于在仿真中成功学习也是必要的。

由于模拟在表达能力上受到这些实际限制，机器人在部署阶段经常会遇到训练期间未建模的场景。例如，如果机器人只在平坦地面上训练，且训练时没有改变地形几何，它可能无法翻越楼梯等非平坦地形。在这种情况下，通常的做法是调整训练环境或奖励函数并重新训练策略。这种“重新训练—真实世界测试”的迭代循环非常繁琐。更糟糕的是，在某些场景下，仅靠这种迭代并不足够，因为无法准确建模或感知重要的环境属性。例如，茂密的灌木丛既难以模拟（因为其柔顺性），又难以感知（因为深度传感器无法将其与墙壁区分）。因此，机器人可能会像攀爬岩石一样试图越过灌木丛，或以过于保守、导致机器人被困的步态试图穿过灌木丛。

上述例子表明，即便对于最先进的仿真到现实系统，现实世界仍会带来新的挑战。我们将那些可以在仿真中模拟但在训练中未被预料到的场景，以及那些无法从仿真中建模或无法从感觉观测中识别的情况，统称为分布外情况。我们提出一个策略学习框架，能够在某些假设下提升在分布外场景中的表现。我们的关键洞见是：给定一个任务，存在多个同等优秀的解（即欠规范[4]），它们在训练表现上相当，但可以以不同方式泛化。例如，在平坦地面上行走这一任务只约束机器人身体的速度，而不约束腿部如何运动、躯干应离地多高等等。考虑两种不同的行走行为：crouch（蹲伏行走）让机器人躯干贴近地面，stomp（高抬步行走）则保持躯干较高且腿部有较高的抬脚幅度。虽然crouch和stomp都能在平坦路面上成功行走，但它们在分布外场景中的泛化能力不同：采用crouch时，机器人可以从障碍物下方穿过，但无法爬楼梯；而采用stomp时，机器人可以翻越路沿/楼梯，但无法从障碍物下方穿过（图1）。

在训练环境中，虽然有许多可能的运动行为都能成功，但典型的强化学习建模方式最终只会产生一种策略，该策略只体现一种解决方案，因而具有单一的泛化偏差。为了促进对多样化场景的泛化，我们提出了一种名为“行为多样性”（Multiplicity of Behavior, MoB）的技术，该技术在给定相同的观察历史以及一小组行为参数时，能输出不同的行走行为。当面对未见过的场景时，可以通过改变这些参数来测试不同的行为，这比重新训练一个策略要快得多，并且便于人类操作员收集在线演示。

MoB的效用依赖于一个假设：在受限训练环境中成功的一些行为子集，也将能在分布外的目标环境中成功。为了证明这一假设在实质意义上成立，我们选择了一个极端场景：仅在地平面上训练一个四足行走策略，然后在非平坦地形和新任务上进行评估。我们展示了人类操作员可以实时调整行为，以便在多种未见的地形和动力学条件下实现成功运动，包括不平整地面、楼梯、外部推挤和受限空间（图1）。同样的调整机制还可用于组合行为以执行新任务，例如载荷操作和有节奏的舞蹈（图1）。

这项工作贡献了一个稳健的低层四足控制器，能够执行多样化的结构化行为，我们希望它将成为未来运动控制应用中有用的构建模块。它可以作为一个平台，通过可解释的高层控制接口收集各种任务的四足演示。此外，我们的控制器展示了 MoB 作为分布外泛化实用工具的价值。虽然我们的 MoB 实现利用了运动控制领域的专家知识，但总体而言，学习多种达成目标的方法以促进泛化，这一技术是一种有前景的路径，在机器人领域从仿真到现实的强化学习中具有广阔的应用潜力。

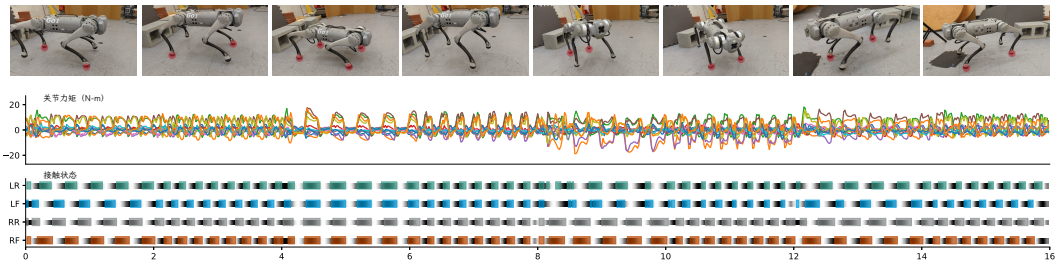


图2：一个学习到的控制器在经典结构化步态之间切换：小跑、弹跳、踱步和原地跳跃，以2Hz和4Hz交替频率进行。图像显示了机器人实现每种步态接触相位的过程，支撑脚以红色突出显示。底部绘图的黑色阴影反映了每只脚的时序参考变量t_t；彩色条报告由足部传感器测量的接触状态。正如我们后面将要揭示的，多样化行为可以促进新颖的下游能力。

2 背景

辅助奖励。仅使用任务奖励（例如速度跟踪）学习到的运动步态尚未被证明能够成功迁移到现实世界。因此必须使用辅助奖励进行训练，这些奖励引导机器人保持特定的接触时序、动作平滑度、能耗或足部离地高度 [1, 2, 3, 5, 6, 7],，以补偿仿真到现实的差距。这类辅助奖励可以理解为用于泛化的偏置。例如，足部离地高度能使机器人在现实世界地形更不平整，或机器人在现实世界中的下陷幅度比仿真中更大时依然保持鲁棒 [7]。当智能体在现实任务中失败时，常见做法是手动调整辅助奖励，以促使成功现实行为的出现。然而，这种调整很繁琐，因为它需要反复迭代训练和部署。此外，这种调整是任务特定的，每当向智能体下达新任务时都必须重新进行。设计一套能在多种下游任务中促进泛化的辅助奖励的难度，如图 1 顶部插图所示：每张插图都展示了一个实例，其中某个固定的辅助奖励对一项任务有效，却会导致另一项任务失败。

参数化风格的运动学习。最近与双足机器人密切相关的相关工作，已通过参数增强的辅助奖励项，将步态参数明确纳入任务规范中，包括对接触模式 [8] 和目标足部放置位置 [9]的追踪奖励。 [8]在双足运动中使用依赖于步态的奖励项，以控制两足之间的时间偏移和摆动阶段的持续时间。 [8] 的奖励结构启发了我们，但由于其命令参数空间较小，并未提出也无法支持在组合任务或分布外环境中的应用。类似地，其他工作通过参考步态库施加参数化风格。 [10, 11] 生成了一个参考轨迹库，并训练了一个目标条件策略来模仿这些轨迹。 [12] 证明了可以从参考轨迹库中同时学习一小部分离散的运动风格。

多样性目标学习。若干先前方法旨在自动学习一组高性能且多样的行为。质量多样性（QD）方法 [13, 14, 15]通过在轨迹之间定义的新颖性目标来学习一个多样化的策略库。它们通常使用进化策略来执行这种优化。QD 已展现出诸多优势，包括改进优化性能和技能复用以支持在线适应 [13, 14]。另一条研究路线在 RL 中使用无监督目标进行技能发现，以改善优化 [16]或分布外泛化 [17, 18]。无监督多样性方法前景广阔，但尚未扩展到真实机器人平台，也未产生用于指导行为的、可落地的参数接口。

基于步态参数的分层控制。已有若干工作通过学习高层策略，调节底层基于模型的控制器的步态参数，以完成下游任务。该方法此前已应用于能量最小化 [19] 和视觉引导的落脚点 [20, 21]。这些工作依赖模型预测的底层控制器来执行不同步态，而没有使用步态条件化的学习控制器。模型控制与强化学习之间的权衡已在先前文献中得到了充分讨论 [3, 22]。我们的工作使得在高层和底层都进行学习的分层方法能够被重新审视。

Term	方程	权重
$r_{v_{x,y}^{\text{cmd}}}$：xy 速度跟踪	$\exp\{-\lvert v_{xy} - v_{xy}^{\text{cmd}}\rvert^2/\sigma_{v_{xy}}\}$	0.02
$r_{\omega_z^{\text{cmd}}}$：偏航角速度跟踪	$\exp\{-(\omega_z - \omega_z^{\text{cmd}})^2/\sigma_{\omega_z}\}$	0.01
$r_{c_f^{\text{cmd}}}$：摆动相跟踪（力）	$\sum_{\text{足}}[1 - C^{\text{cmd}}_{\text{足}}(\theta^{\text{cmd}}, t)]\exp\{-\lvert f^{\text{foot}}\rvert^2/\sigma_{cf}\}$	−0.08
$r_{c_v^{\text{cmd}}}$：支撑相跟踪（速度）	$\sum_{\text{foot}}[C^{\text{cmd}}_{\text{foot}}(\theta^{\text{cmd}}, t)]\exp\{-\lvert v_{xy}^{\text{foot}}\rvert^2/\sigma_{cv}\}$	−0.08
$r_{h_z^{\text{cmd}}}$：身体高度跟踪	$(h_z - h_z^{\text{cmd}})^2$	−0.2
r_{ϕ}命令：机体俯仰跟踪	$(\phi - \phi^{\text{cmd}})^2$	−0.1
$r_{s_y^{\text{cmd}}}$：Raibert 启发式足部摆动跟踪	$(p^f_{x,y\text{足端}} - p^f_{x,y\text{foot}}(s_y^{\text{cmd}}))^2$	−0.2
$r_{h_z^f}$指令：足端摆动高度跟踪	$\sum_{\text{脚}}(h^f_{z,\text{脚}} - h^f_{z\text{命令}})^2 C^{\text{cmd}}_{\text{脚}}(\theta^{\text{cmd}}, t)$	−0.6
z 轴速度	v_z^2	−4e−4
横滚-俯仰角速度	$\lvert\omega_{xy}\rvert^2$	−2e−5
足部打滑	$\lvert v_{xy}^{\text{foot}}\rvert^2$	−8e−4
大腿/小腿碰撞	$\mathbb{1}_{\text{collision}}$	−0.02
关节限位超出	$\mathbb{1}_{q_i > q_{max} \lvert\rvert q_i < q_{min}}$	−0.2
关节力矩	$\lvert\tau\rvert^2$	−2e−5
关节速度	$\lvert\dot{q}\rvert^2$	−2e−5
关节加速度	$\lvert\ddot{q}\rvert^2$	−5e−9
动作平滑	$\lvert a_{t-1} - a_t\rvert^2$	−2e−3
动作平滑，二阶	$\lvert a_{t-2} - 2a_{t-1} + a_t\rvert^2$	−2e−3

Task

增强辅助

固定辅助

Table 1：奖励结构：任务奖励、增强辅助奖励和固定辅助奖励 .

3 方法

为了获得 MoB，我们训练了一个条件策略 $\pi(\cdot|c_t, b_t)$，它能够以多种方式完成由命令 (c_t) 指定的任务，这些方式源于对行为参数 b_t 的不同选择。于是问题来了：如何定义 b_t。我们可以使用无监督的多样性度量来学习行为，但这些行为可能并不实用 [4]，而且无法由人工调节。为了解决这些问题，我们利用人类对有用行为参数 (b_t) 的直觉，这些参数对应步态特性，例如足部摆动运动、身体姿态和接触时序 [6, 7, 8, 23]。在训练过程中，智能体接收任务奖励（用于速度跟踪）、固定辅助奖励（用于促进仿真到现实迁移和稳定运动），以及增强辅助奖励（用于鼓励以期望风格运动）的组合。在部署到新环境时，人类操作员可以通过改变策略的输入 b_t 来调节策略的行为。

3.1 MoB 的任务结构

任务规范。我们考虑全向速度跟踪任务。该任务由一个三维命令向量 $c_t = [v_x^{\text{cmd}}, v_y^{\text{cmd}}, \omega_z^{\text{cmd}}]$ 指定，其中 v_x^{cmd}, v_y^{cmd} 是机体坐标系 x 轴和 y 轴上的期望线速度，ω_z^{cmd} 是偏航轴上的期望角速度。

Beh行为规范。我们通过一个 8 维向量来参数化任务完成的风格。 r
of b行为参数 b_t:

$$\mathbf{b}_t = [\theta_1^{\text{cmd}}, \theta_2^{\text{cmd}}, \theta_3^{\text{cmd}}, f^{\text{cmd}}, h_z^{\text{cmd}}, \phi^{\text{cmd}}, s_y^{\text{cmd}}, h_z^{f\,\text{cmd}}].$$

$\theta^{\text{cmd}} = (\theta_1^{\text{cmd}}, \theta_2^{\text{cmd}}, \theta_3^{\text{cmd}})$ 是成对足部之间的时间偏移。这些偏移表达步态，包括弹跳步态 $(\theta^{\text{cmd}} = (0.0, 0, 0))$、小跑步态 $(\theta^{\text{cmd}} = (0.5, 0, 0))$、跳跃步态 $(\theta^{\text{cmd}} = (0, 0.5, 0))$、同侧步态 $(\theta^{\text{cmd}} = (0, 0, 0.5))$，以及它们的连续插值，例如疾驰步态 $(\theta^{\text{cmd}} = (0.25, 0, 0))$。总的来说，参数 θ^{cmd} 可以表达所有两拍四足接触模式；图 2 提供了直观图示。f^{cmd} 是以 Hz 表示的步频。例如，指令 $f^{\text{cmd}} = 3$ Hz 将导致每只脚每秒接触地面三次。h_z^{cmd} 是身体高度指令；ϕ^{cmd} 是身体俯仰指令。s_y^{cmd} 是足部站位宽度指令；$h_z^{f\,\text{cmd}}$ 是足部摆动高度指令。

奖励函数。所有奖励项列于表1。身体速度跟踪的任务奖励定义为命令向量c_t的函数。辅助奖励用于约束

四足动物的运动出于各种原因。“固定”辅助奖励与行为参数(b_t)无关，并促进所有步态的稳定性和平滑度，以实现更好的仿真到现实迁移。训练中，一个担忧是当任务奖励被辅助目标的惩罚所压倒时，机器人可能会放弃其任务或选择提前终止。为了解决这个问题，如 [7], 中那样，我们通过将总奖励计算为 $r_{\text{task}} \exp(c_{\text{aux}} r_{\text{aux}})$，强制总奖励成为任务奖励的正线性函数，其中 r_{task} 是（正）任务奖励项的总和，r_{aux} 是（负）辅助奖励项的总和（我们使用 $c_{\text{aux}} = 0.02$）。这样，智能体总是因其朝着任务前进而获得奖励，当辅助目标得到满足时奖励更多，而当它们未得到满足时奖励更少。

对于 MoB，我们将增强的辅助奖励定义为行为向量 b_t 的函数。我们设计这些奖励，使其在实现的行为与 b_t 匹配时增加，并且不与任务奖励冲突。这需要对奖励结构进行仔细设计。例如，当将支撑宽度实现为行为参数时，一种简单的方法是直接奖励左右脚之间恒定的期望距离。然而，这会在需要脚部相对横向运动的快速转弯任务中惩罚机器人。为了避免这种情况，我们采用了 Raibert 启发式方法，它建议为实现特定的身体速度和接触时序 [23, 24] 所需的脚部运动学运动。Raibert 启发式方法计算地面平面中的期望脚位置 $\boldsymbol{p}^{f}_{x,y\,\text{foot}}^{\text{cmd}}(\boldsymbol{s}^{\text{cmd}}_{y})$，作为对基准支撑宽度的调整，使其与期望的接触时序和身体速度一致。为了定义期望的接触时序，函数$C^{\text{cmd}}_{\text{foot}}(\boldsymbol{\theta}^{\text{cmd}}, t)$ 根据相位和时间变量计算每只脚的期望接触状态，如 [8], 所述，详细内容见附录。

3.2 学习多样化运动

任务与行为采样。为了学习行为之间优雅的在线过渡，我们在每个训练回合内重新采样所需的任务和行为。为了使机器人既能快速奔跑又能快速旋转，我们使用来自 [3]的网格自适应课程策略对任务 $c_t = (v^{\text{cmd}}_x, v^{\text{cmd}}_y, \boldsymbol{\omega}^{\text{cmd}}_z)$ 进行采样。然后，我们需要采样一个目标行为 b_t。首先，我们采样 $(\boldsymbol{\theta}^{\text{cmd}}_1, \boldsymbol{\theta}^{\text{cmd}}_2, \boldsymbol{\theta}^{\text{cmd}}_3)$ 作为对称四足接触模式之一（腾跳、小跑、跳跃或侧步），这些模式被认为更稳定，并且我们发现它们足以构成多种有用步态的基础。然后，其余命令参数 $(v^{\text{cmd}}_y, \boldsymbol{f}^{\text{cmd}}, \boldsymbol{h}^{\text{cmd}}_z, \boldsymbol{\phi}^{\text{cmd}}, \boldsymbol{h}^{f}_z \text{cmd}\, \boldsymbol{s}^{\text{cmd}}_y)$ 独立且均匀地采样。它们的取值范围见表 6。

策略输入。策略的输入是观测 $o_{t-H\ldots t}$、命令 $c_{t-H\ldots t}$、行为 $b_{t-H\ldots t}$、先前动作 $a_{t-H-1\ldots t-1}$ 和时序参考变量 $t_{t-H\ldots t}$ 的 30 步历史。观测空间 o_t 由关节位置和速度 $q_t, \dot{q}_t$ （由关节编码器测量）以及机体坐标系中的重力向量 g_t （由加速度计测量）组成。时序参考变量 $t_t = [\sin(2\pi t^{\text{FR}}), \sin(2\pi t^{\text{FL}}), \sin(2\pi t^{\text{RR}}), \sin(2\pi t^{\text{RL}})]$ 由每只脚的偏移时间计算得出：
$[t^{\text{FR}}, t^{\text{FL}}, t^{\text{RR}}, t^{\text{RL}}] = [t + \boldsymbol{\theta}^{\text{cmd}}_2 + \boldsymbol{\theta}^{\text{cmd}}_3, t + \boldsymbol{\theta}^{\text{cmd}}_1 + \boldsymbol{\theta}^{\text{cmd}}_3, t + \boldsymbol{\theta}^{\text{cmd}}_1, t + \boldsymbol{\theta}^{\text{cmd}}_2]$，其中 t 是一个计数器变量，在每个步态周期中从 0 递增到 1，FR、FL、RR、RL 分别为四只脚。这种形式改编自 [8]，用于表达四足步态。
策略架构。我们的策略主体是一个MLP，隐藏层大小为 $[512, 256, 128]$，并使用ELU激活函数。除此之外，策略输入还包括估计的域参数：机器人身体的速度和地面摩擦力，这些参数通过监督学习从观测历史中预测，方式同 [7]。估计器模块是一个MLP，隐藏层大小为 $[256, 128]$，并使用ELU激活函数。我们没有分析该估计对性能的影响，但发现它有助于可视化部署。

动作空间。动作a_t 由十二个关节各自的位置目标组成。零动作对应名义关节位置$\hat{}q$。位置目标通过比例-微分控制器进行跟踪，其中参数为 $k_p = 20$、 $k_d = 0.5$。

3.3 面向仿真到现实迁移的设计选择

域随机化。为了更好的仿真到现实迁移，我们训练一个策略，使其对一系列机器人身体质量、电机强度、关节位置标定、地面摩擦力和恢复系数、重力的方向和大小具有鲁棒性。由于我们关注分布外泛化研究，我们仅在平坦地面上训练，不对地形几何做任何随机化。这一选择也简化了训练。所有参数的随机化范围见附录A。

Gait	0.0 m/s	1.0 m/s	2.0 m/s	3.0 m/s
Trotting	$9_{\pm 1}$	$24_{\pm 1}$	$53_{\pm 5}$	$98_{\pm 9}$
Pronking	$32_{\pm 1}$	$43_{\pm 2}$	$68_{\pm 5}$	$112_{\pm 5}$
Pacing	$13_{\pm 3}$	$25_{\pm 2}$	$55_{\pm 3}$	$99_{\pm 6}$
Bounding	$22_{\pm 2}$	$39_{\pm 4}$	$78_{\pm 5}$	$127_{\pm 35}$
Gait-free Baseline	$17_{\pm 5}$	$35_{\pm 5}$	$64_{\pm 10}$	$102_{\pm 14}$
Trotting ($\boldsymbol{f}^{\text{cmd}} = 2$ Hz)	$11_{\pm 2}$	$25_{\pm 1}$	$55_{\pm 4}$	$104_{\pm 8}$
Trotting ($\boldsymbol{f}^{\text{cmd}} = 3$ Hz)	$9_{\pm 1}$	$24_{\pm 1}$	$53_{\pm 5}$	$98_{\pm 9}$
Trotting ($\boldsymbol{f}^{\text{cmd}} = 4$ Hz)	$9_{\pm 1}$	$26_{\pm 0}$	$60_{\pm 4}$	$114_{\pm 12}$
Trotting ($\boldsymbol{h}_z^{\text{cmd}} = 20$ cm)	$9_{\pm 1}$	$26_{\pm 1}$	$56_{\pm 3}$	$102_{\pm 8}$
Trotting ($\boldsymbol{h}_z^{\text{cmd}} = 30$ cm)	$9_{\pm 1}$	$24_{\pm 1}$	$53_{\pm 5}$	$98_{\pm 9}$
Trotting ($\boldsymbol{h}_z^{\text{cmd}} = 40$ cm)	$10_{\pm 1}$	$23_{\pm 1}$	$52_{\pm 4}$	$95_{\pm 9}$

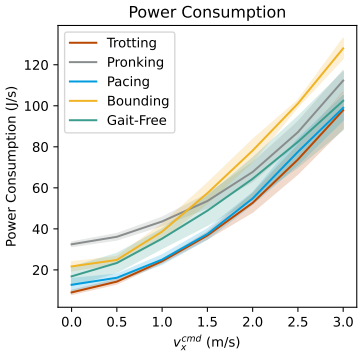


Table 3: Behavior tuning enables interventional studies on the relationship between gait properties and performance criteria within a single policy. Here, we illustrate how power consumption varies across speeds for common quadrupedal gaits and for a baseline policy without gait constraint. Several structured gaits surpass the efficiency of unconstrained gait across all speeds.

Latency and Actuator Modeling. Directly identifying invariant properties avoids overly conservative behavior from unnecessary domain randomization. We perform system identification to reduce the sim-to-real gap in the robot dynamics. Following [22], we train an actuator network to capture the non-ideal relationship between PD error and realized torque. Separately, we identify a latency of around 20 ms in our system and model this as a constant action delay during simulation.

3.4 Materials

Simulator and Learning Algorithm. We define our training environment in the Isaac Gym simulator [6, 25]. We train policies using Proximal Policy Optimization [26]; details in Appendix A.

Hardware. We deploy our controller in the real world on the Unitree Go1 Edu robot [27]. An onboard Jetson TX2 NX computer runs our trained policy. We implement an interface based on Lightweight Communications and Marshalling (LCM) [28] to pass sensor data, motor commands, and joystick state between our code and the low-level control SDK provided by Unitree. For both training and deployment, the control frequency is 50Hz.

Gait-free Baseline. To understand the impact of MoB on performance, we compare our controller to a baseline velocity-tracking controller (the "gait-free baseline"). The gait-free baseline is trained by the method above, but excludes all augmented auxiliary rewards (Table 1). Therefore, it only learns one solution to the training environment and its actions are independent of behavior parameters $\mathbf{b}_t$.

4 Experimental Results

4.1 Sim-to-Real Transfer and Gait Switching

We deploy the controller learned in simulation in the real world and first evaluate its performance on flat ground similar to the training environment. To start, we demonstrate generating and switching between structured gaits that are well-known in the locomotion community. Figure 2 shows torques and contact states during transition between trotting, pronking, bounding, and pacing while alternating $\boldsymbol{f}^{\text{cmd}}$ between 2 Hz and 4 Hz. We find that all gait parameters are consistently tracked after sim-to-real transfer. Videos (i)-(iv) on the project website visualize the different gaits obtained by modulating each parameter in $\mathbf{b}_t$ individually.

4.2 Leveraging MoB for Generalization

Tuning for New Tasks. After training using a generic locomotion objective, one might wish to tune a controller's behavior to optimize a new metric in the original environment. MoB facilitates this if some subset of learned behaviors outperform the gait-free policy by the new task metric.

Gait	$r_{v^{\text{cmd}}_{x,y}}$	$r_{\omega^{\text{cmd}}_z}$	$r_{c^{\text{cmd}}_f}$	$r_{c^{\text{cmd}}_v}$	Survival
Trotting	$0.80^{(0.95)}_{\pm 0.01}$	$0.76^{(0.89)}_{\pm 0.00}$	$0.95^{(0.97)}_{\pm 0.00}$	$0.98^{(0.98)}_{\pm 0.00}$	$0.88^{(1.00)}_{\pm 0.01}$
Pronking	$0.84^{(0.94)}_{\pm 0.01}$	$0.77^{(0.85)}_{\pm 0.01}$	$0.96^{(0.96)}_{\pm 0.00}$	$0.97^{(0.98)}_{\pm 0.00}$	$0.82^{(1.00)}_{\pm 0.02}$
Pacing	$0.76^{(0.91)}_{\pm 0.01}$	$0.76^{(0.81)}_{\pm 0.01}$	$0.94^{(0.96)}_{\pm 0.00}$	$0.98^{(0.98)}_{\pm 0.00}$	$0.87^{(1.00)}_{\pm 0.02}$
Bounding	$0.80^{(0.88)}_{\pm 0.01}$	$0.73^{(0.86)}_{\pm 0.01}$	$0.94^{(0.96)}_{\pm 0.00}$	$0.98^{(0.98)}_{\pm 0.00}$	$0.82^{(1.00)}_{\pm 0.01}$
Gait-free	$0.81^{(0.96)}_{\pm 0.03}$	$0.74^{(0.92)}_{\pm 0.06}$	–	–	$0.83^{(1.00)}_{\pm 0.01}$

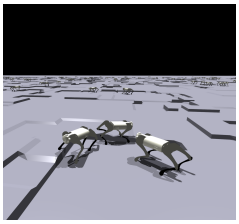


Table 4: Zero-shot generalization to platform terrain (visualized right). Pacing and trotting yield the best survival time in out-of-distribution deployment, outperforming the gait-free baseline. Pronking attains the best velocity tracking performance, with similar survival time to the baseline. We report the fraction of maximum episodic reward. Superscript reports performance in flat training environment with no platforms. Subscript reports standard deviation across three random seeds.

Energy efficiency (simulated): We consider the task of minimizing the mechanical power consumption ($\mathrm{J/s}$) measured by summing the product of joint velocity and torque at each of the 12 motors: $\sum_i \max(\boldsymbol{\tau}_i \dot{\mathbf{q}}_i, 0)$. As shown in Figure 3, several choices of the contact schedule $\boldsymbol{\theta}^{\text{cmd}}$, height $\boldsymbol{h}_z^{\text{cmd}}$, and frequency $\boldsymbol{f}^{\text{cmd}}$ outperform the gait-free policy in this unseen metric, consuming less energy across all speeds. Therefore, tuning the behavior parameters can facilitate tuning performance on a new objective (energy efficiency) in the original training environment.

Payload manipulation: We experiment with another task where the robot is required to transport a ball from one place to another, then bend its body so the ball is deposited on the ground. A gait-free policy couldn't do this; while many possible body posture profiles are valid to solve the training environment, the gait-free policy will simply converge to one at random. In a real-world experiment, we demonstrate how MoB-enabled body posture control can be repurposed for teleoperated payload manipulation. The operator pilots the robot to the delivery location with the body level, then pitches the body backward, modulating ϕ^{cmd}, to dump the payload (Figure 1, bottom row, second from left).

Tuning for New Environments. In the previous section, we showed that MoB can support repurposement to novel tasks in the training environment. In the real world, there is also always a long tail of environments that are not modeled in training. We demonstrate the behavior of our controller trained *only on flat ground* in the presence of challenging non-flat terrains and disturbances such as curbs, bushes, hanging obstacles, and shoves. Successes suggest that the space of policies learned by MoB contains some behaviors that transfer better than the baseline to unseen environments.

Platform terrain (simulated): We evaluate the performance of our robot in traversing randomly positioned platforms with height up to $16\,\mathrm{cm}$ (Table 4). We report two metrics: mean reward and mean survival time as a fraction of the maximum episode length ($10\,\mathrm{s}$). For each metric, we find that by modulating the contact schedule $\boldsymbol{\theta}^{\text{cmd}}$ or footswing height $\boldsymbol{h}_z^{f\,\text{cmd}}$, we can outperform the gait-free policy. (Figure 4, Appendix Figure 9). Therefore, it is possible to improve performance in an out-of-distribution terrain by modulating the parameters of the MoB policy.

Climbing over curbs: Previous works demonstrating blind obstacle traversal on a quadruped [1, 2] learned a foot-trapping reflex where the robot first trips, then raises its foot over the obstacle. In contrast, with the help of a human pilot, our gait-conditioned policy with high footswing command enables fast and smooth obstacle traversal without tripping, despite training on a simpler flat-ground terrain. In the real world, we demonstrate that modulating footswing height $\boldsymbol{h}_z^{f\,\text{cmd}}$ enables our controller to climb smoothly across stairs and curbs (Figure 1, bottom left).

Hacking through thick bushes: Extremely thick bushes pose a methodological challenge for state-of-the-art perceptive locomotion controllers. They are hard to simulate, because they comply against the whole body of the robot, and they are hard to sense as they are indistinguishable from solid obstacles to depth sensors. Therefore, prior works would either attempt to climb over bushes as obstacles or fall back on a robust proprioceptive controller that is unaware of the semantic context. In contrast, by modulating the gait parameters for gait frequency $\boldsymbol{f}^{\text{cmd}}$ and footswing height $\boldsymbol{h}_z^{f\,\text{cmd}}$ *in situ*, a human operator can guide our system quickly through challenging brush.

Navigating confined spaces: Consider the scenario where the robot needs to go under a bar. The gait-free baseline cannot accomplish this; in the absence of such constraints during training, it will

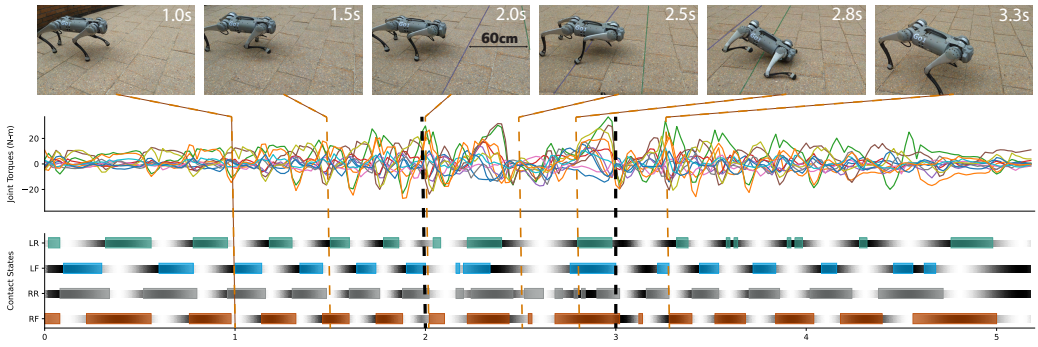


Figure 3: We demonstrate that behavior transitions can be performed even in quick sequence at high speed for synthesis of agile maneuvers. Emulating gap crossing on flat ground, we show that this can facilitate crossing a gap wider than the robot's body length in a single leap.

converge to a fixed body height profile. Our system with MoB can navigate confined spaces through modulation of the body height $\boldsymbol{h}_z^{\text{cmd}}$. In a real-world example, the robot was able to crawl under a $22\,\text{cm}$ bar; the robot body thickness is $13\,\text{cm}$, leaving $9\,\text{cm}$ of clearance beneath the robot.

Anticipative bracing against shoves: Widening the stance can make a quadruped more robust to hard shoves, but absent a perception module to anticipate a human kick, the robot would always need to walk in a widened stance to be prepared for a shove. This interferes with performance in other tasks like running efficiently, so learned locomotion controllers without MoB often provide incentive to keep the feet nominally below the hips [2, 7]. With MoB, widening the stance width $\boldsymbol{s}_y^{\text{cmd}}$ in the pilot's anticipation of a shove enables the robot to remain upright.

Exploring the Speed and Range of Behavior Adaptation. If we want to use our low-level controller for high-level tasks, one thing we might desire is to switch between gaits even at high speeds, which might be useful for applications such as parkour. Another is to transition between diverse gaits with precise timing for a synchronous task like dancing.

Agile forward leap: As a demonstration of gait transitions at high speed, we modulate contact schedule, velocity, and gait frequency at to encode an agile forward leap (Figure 3). The robot first accelerates to a target speed of $3\,\text{m/s}$ at a trot while increasing its step frequency from $2\,\text{Hz}$ to $4\,\text{Hz}$, then switches to pronking at $2\,\text{Hz}$ for one second, then decelerates to a standstill while trotting. During the leap phase, the distance from the location of the robot's front feet at takeoff to the location of the hind feet upon landing is $60\,\text{cm}$.

Choreographed dance: To demonstrate precisely timed transition between diverse gaits, we program a sequence of gait parameters to generate a dance routine synchronized to a jazz song with a tempo of 90 bpm. At this tempo, combinations of phases 0, 0.25, and 0.5 with frequencies of $1.5\,\text{Hz}$ and $3\,\text{Hz}$ yield eighth, quarter, half, and full beat gaps between consecutive footsteps. We also modulate body height and velocity in time with the music. An assistant script procedurally generates gait parameters, fed into the controller in open-loop fashion.

5 Discussion and Limitations

Our experiments show that the benefits of adding MoB can come at a cost to in-distribution task performance, specifically limiting the robot's flat-ground sprinting performance (Table 5, appendix). Heat maps reveal that our behavior parameterization is restrictive for combinations of high linear and angular velocity. To quantify and control the tradeoff between task performance and reward shaping is an interesting future direction, for which some prior methods have been proposed [29].

MoB confers a single learned policy a structured and controllable space of diverse locomotion behaviors for each state and task in the training distribution. This yields a set of 'knobs' to tune the performance of motor skills in unseen test environments. The system currently requires a human pilot to manually tune its behavior. In the future, the autonomy of our system could be extended by automating behavior selection using imitation from real-world human demonstrations, or by using a hierarchical learning approach to automatically self-tune the controller during deployment.

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Author Contributions

- **Gabriel B. Margolis** concieved, designed, and implemented the controller, ran all experiments, and played the primary role in paper writing.
- **Pulkit Agrawal** advised the project and contributed to its conceptual development, experimental design, positioning, and writing.

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